NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

Tests of Immune Function

Shil Punatar, DO
Gastroenterology Fellow, PGY4
Franciscan Health, IL
shilpunatar@gmail.com

and

Tony A. Slieman, M.Sc., Ph.D.
Director of Academic Affairs and Associate Professor
Founding Director of Masters in Biomedical Sciences
Department of Biomedical and Anatomical Sciences
tslieman@nyit.edu

Do.
Make.
Heal.
Innovate.
Reinvent the Future.

Session Objectives

1. Be able to classify which immune function tests are appropriate for each relevant immune cell type
2. Be able to describe the mechanism of how each immune function test works
3. Be able to compare and contrast screening from confirmatory testing
4. Be able to interpret immune function laboratory data findings
5. Be able to differentiate which immune function tests are appropriate for specific disease pathologies

Source: Course Syllabus

What to do first?

#1: ***<u>HISTORY AND PHYSICAL ARE INCREDIBLY IMPORTANT</u>***

1. CBC W/ Differential
2. Chemistry Panel
3. UA
4. Chest Radiograph
5. Erythrocyte sedimentation rate
6. C-Reactive protein

When to be concerned?

History:

- Indicative of a familial origin
- Reason to believe in abnormal development
- Environmental changes
- Abnormal lab work in past: Anemias, Neutropenia
- Recurrent infections, Risk of HIV
- Patient's sex
- Failure to Thrive
- Fever of unknown origin

Physical Examination:

- Indicative of recurrent infections: *Candida albicans*
- Indicative of splenomegaly
- Multiple systemic features
- Check for lymph nodes

Prevalence of Immunodeficiencies

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

- Antibody Deficiencies: 55%
- Cellular/Humoral Deficiency: 15%
- Congenital defects: 10%
- Immune Dysregulation: 5%
- Autoinflammatory: 5%
- Complement: 4%
- Combined Immunodeficiencies: 3%
- Deficiency in innate immunity: 3%

Normal Lab Values

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

Erythrocyte count	4.6-5.9 million/µL
Hg	M: 12-16 g/dL F: 14-18 g/dL
Hct	M: 42-50% F: 37-47%
MCV	80-98 μm^3
MCH	28-32 pg
MCHC	33-36 g/dL
RDW	≤ 9-14.5%
Platelets	150,000-450,000/mm^3

Leukocyte count	4,000-11,000/μm^3
Segmented Neutrophils	50-70%
Bands	0-5%
Eosinophils	0-3%
Basophils	0-1%
Lymphocytes	30-45%
Monocytes	0-6%

(ABIM,2020)

Normal Lab Values

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

Red blood cell (RBC) count[1]

Men:	4.5–5.5 million RBCs per microliter (mcL) or 4.5–5.5 x 10^{12}/liter (L)
Women:	4.0–5.0 million RBCs per mcL or 4.0–5.0 x 10^{12}/L
Children:	3.8–6.0 million RBCs per mcL or 3.8–6.0 x 10^{12}/L
Newborn:	4.1–6.1 million RBCs per mcL or 4.1–6.1 x 10^{12}/L

Hemoglobin (Hgb)[1]

Men:	14–17.4 grams per deciliter (g/dL) or 140–174 grams per liter (g/L)
Women:	12–16 g/dL or 120–160 g/L
Children:	9.5–20.5 g/dL or 95–205 g/L
Newborn:	14.5–24.5 g/dL or 145–245 g/L

Platelet (thrombocyte) count[1]

Adults:	140,000–400,000 platelets per mm^3 or 140–400 x 10^9/L
Children:	150,000–450,000 platelets per mm^3 or 150–450 x 10^9/L

(Staff, H. 2019)

Acute Phase Reactants

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

- Elevated in both acute & chronic inflammatory states
- Non-specific measure
- Indicative of general status of IS at the time
- Anything that increases by 25% during an inflammatory state
- **C-Reactive Protein (CRP):**
  -PRP
- **Erythrocyte Sedimentation Rate (ESR):**
  -Measures viscosity of blood
  -Increase in viscosity from fibrinogen break down

Acute Phase Reactants

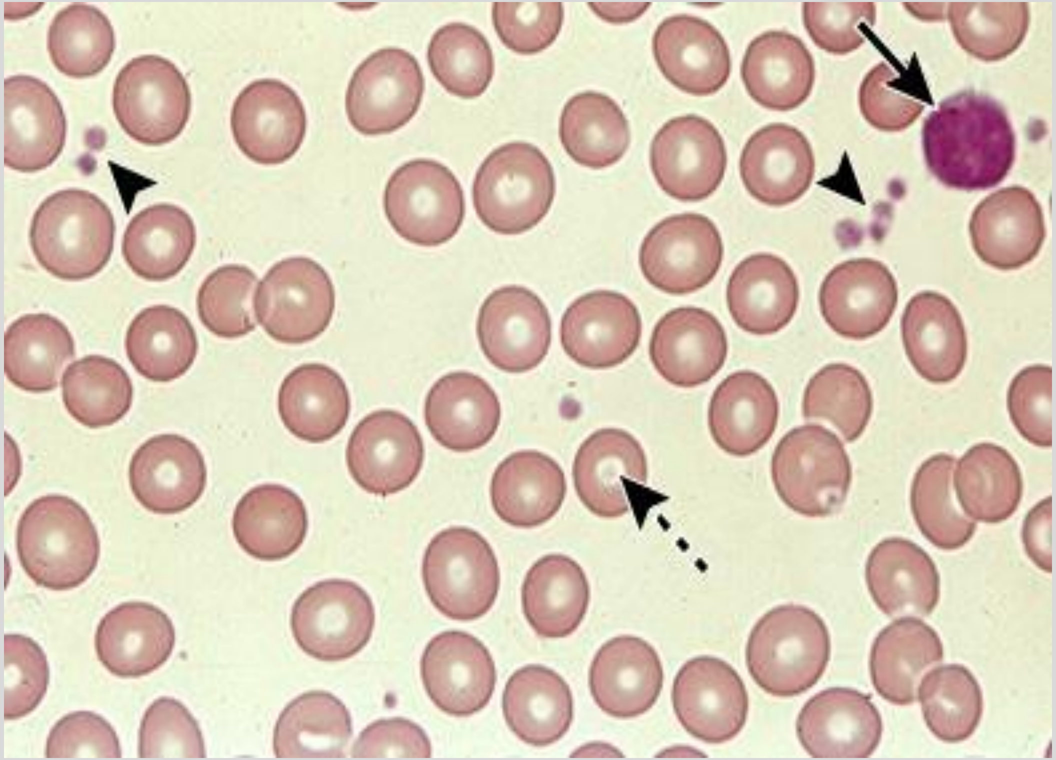
NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

- **C-Reactive Protein (CRP):**
  -PRP
- **Erythrocyte Sedimentation Rate (ESR):**
  -Measures viscosity of blood
  -Increase in viscosity from fibrinogen break down

- Named after reaction to ’C’ arbohydrate proteins from ‘C’apsule of streptococcus

- Elevations in inflammation, infection, injury, occasionally in chronic illness, age determinant

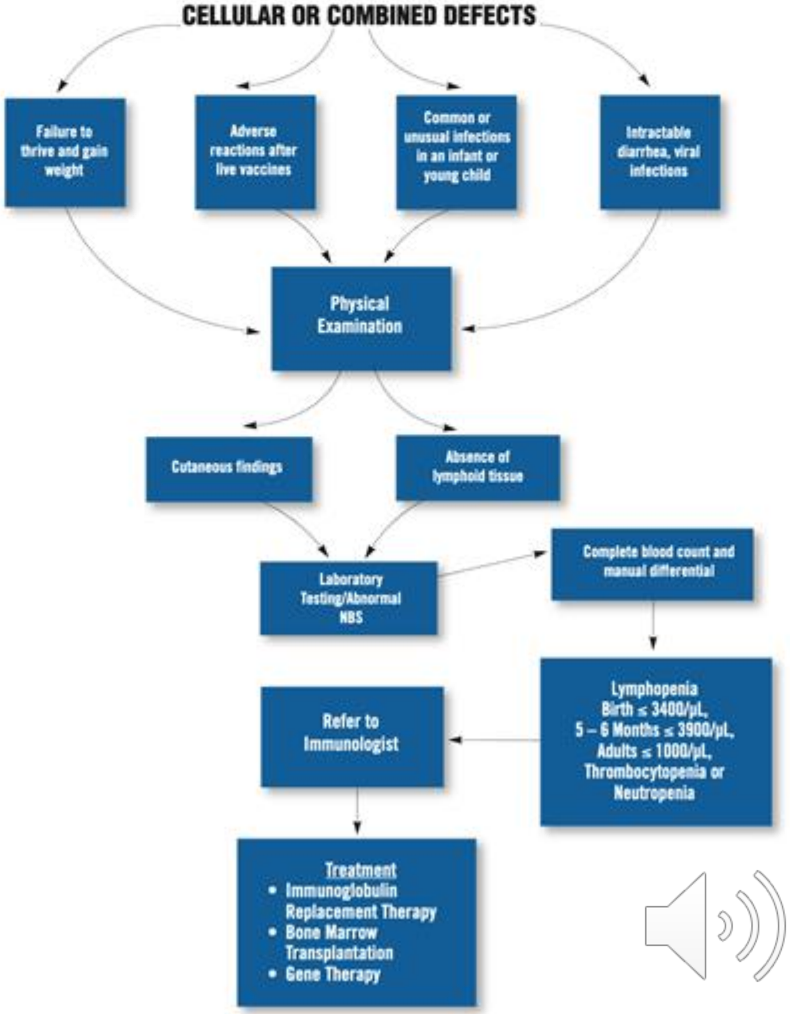
Blood Smear



Visualize:
1. RBC
2. Platelets
3. WBC

Cellular Defects

- Neutrophils: Innate
  - Phagocytosis with bactericidal enzymes
  - Provide extra support to macrophages
  - Derived from bone marrow
- NK cells: Innate
  - targets intracellular pathogens & tumors
  - IFN-Gamma to activate macrophages
- T-cells: Adaptive
  - CD4: regulator of adaptive IR
  - CD8: targets intracellular pathogens & tumors
- B-cells: Adaptive
  - Ab production



(Immune Deficiency Foundation, 2015)

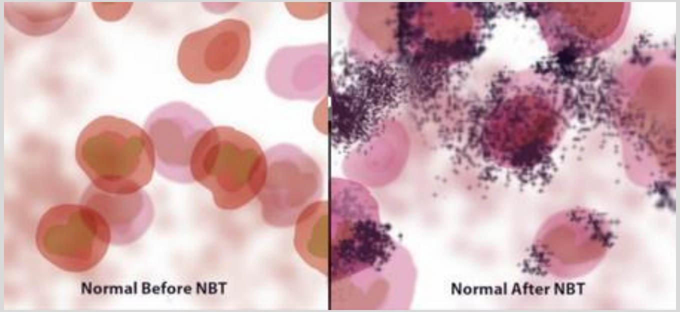
Neutrophil Enumeration & Quality

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

- CBC w/diff & Absolute Neutrophil count:
  - Neutropenia:
    - Congenital: Severe congenital neutropenia
    - Iatrogenic: Cancer treatments
    - Infectious: Viral infections

- Dihydrorhodamine: DHR Flow Cytometry assay
  - -Green fluorescence dye absorbed by cells
  - -Chronic Granulomatous Disease

- Nitroblue Tetrazolium: NBT: NADPH Oxidase activity
  - -Yellow to Blue by NADPH Oxidase
  - -Chronic Granulomatous disease



(Liu, Lucy, 2021)

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

Neutropenia with Fever

- Calculate the Absolute Neutrophil Count
- Will be included on your CBC with differential
- WBC x total neutrophils (bands and segs) x 10
- Severe neutropenia is under 500.
- Concern for infection and possibility for growth stimulating factor initiation

Lymphocyte Enumeration

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

Markers commonly used for assessment of lymphocyte subsets by flow cytometry

Marker name	Cell type	Comment
CD3	T cells	Expressed on all T cells and no other cell type
CD4	T cell subset	Predominantly helper/inducer T cells
CD8	T cell subset	Predominantly cytotoxic T cells; expressed by up to one-third of NK cells
CD19 or CD20	B cells	
CD16	NK cells	Some NK cells may not express CD16
CD56	NK cells	Expressed on the majority of NK cells
CD57	NK cells	Expressed on the majority of NK cells; combinations of CD16, CD56, and CD57 will more reliably evaluate NK cell number than any marker alone
CD45RA	Naive T cells	
CD45RO	Memory T cells	

CD: cluster of differentiation; NK: natural killer.

Graphic 78061 Version 5.0

Causes of lymphocytopenia

Infection	
	Viral (eg, HIV, SARS, SARS-CoV-2, influenza, measles, hepatitis)
	Bacterial (eg, tuberculosis, typhoid fever, brucellosis)
	Fungal (eg, histoplasmosis)
	Parasitic (eg, malaria)
Congenital immunodeficiency disorders	
Iatrogenic	
	Immunosuppressive agents (eg, glucocorticoids, antilymphocyte globulin, alemtuzumab, rituximab)
	Chemotherapy (eg, fludarabine, cladribine), hematopoietic cell transplantation
	Radiation therapy (eg, total body irradiation, radiation injury)
Systemic disease	
	Autoimmune disorders (eg, systemic lupus erythematosus, rheumatoid arthritis, Sjögren syndrome)
	Lymphoma
	Other malignancies
	Sarcoidosis
	Renal failure
	Aplastic anemia, pancytopenia
	Cushing's syndrome
Other causes	
	Alcohol abuse
	Zinc deficiency
	Malnutrition, stress, exercise, trauma
	Thoracic duct leak, rupture, diversion
	Protein-losing enteropathy
	Idiopathic CD4+ lymphocytopenia

HIV: Human immunodeficiency virus; SARS: Severe acute respiratory syndrome; SARS-CoV-2: SARS coronavirus 2.

Screening Tests:

- HIV: CD4+ ↓
- Thymic Aplasia: CD4+, CD8+ ↓
  DiGeorge & Velocardiofacial syndrome
- Severe Combined Immunodeficiency: CD4+, CD8+ ↓
- Ataxia Telangiectasia: CD4+, CD8+ ↓

Lymphocyte Adhesion Defects (LAD)

- Integrin defects: Cells not able to migrate into the periphery tissue.

- 1st sign is **Lymphocytosis**: increased lymphocytes in CBC

1. LAD1: β2 **Integrin** defect
   - $CD_{11a}CD_{18}$ (LFA)
   - $CD_{11b}CD_{18}$ (Mac-1)
   - $CD_{11c}CD_{18}$ (p150,95)
2. LAD2: Fucosylated carbohydrate of **Selectin** is absent
3. LAD3: β1, β2, & β3 **integrins** defective

Lymphocyte Enumeration: **NK cells**

- Check Levels for CD16+/CD56+:
  - -NK deficiency syndrome: No NK cells
  - -SCID: T, B, & NK cells
  - -Paroxysmal nocturnal hemoglobinuria: Pancytopenia
  - -Ataxia Telangiectasia: ↑
- Consider testing in severe, recurrent, atypical HSV infection
- Deficiency leads to susceptibility to: VZV, HSV, EBV, & CMV

Lymphocyte Enumeration: **T-cells**

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

-Incredibly important for T-cell function & level to be intact

-Can lead to several combined pictures of immunodeficiency

Check Levels for:

- CD2+
- CD3+
- CD4+ & CD8+

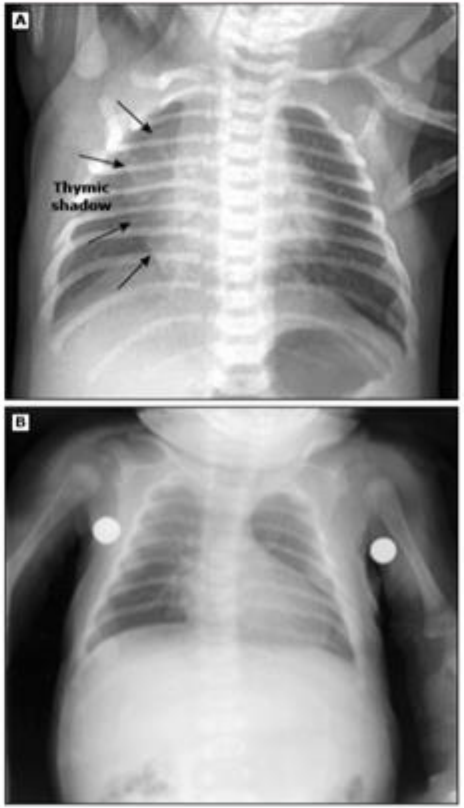
  -Thymic Aplasia: secondary to DiGeorge's Syndrome

  -SCID: T, B, & NK cells

  -Ataxia-Telangiectasia: ATM gene

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

Chest radiographs with and without thymic shadows



The top chest radiograph (A) shows a normal thymic shadow. The bottom radiograph (B) shows absence of the thymic shadow in an infant with severe combined immunodeficiency (SCID).

Courtesy of J Pierre Sasson, MD.

HIV

- Screening:
  - -ELISA: P24: Surface Antigen
  - -IgM, IgG
- Confirmatory:
  - Nucleic acid amplification test
- PCR: qPCR of viral RNA
- Chronic HIV infection w/o treatment: Measure **CD4+ cells**

Lymphocyte Function: **T-cells**

- Stimulation of T-cells:
  - PHA: Phytohemagglutinin
    - -Low stimulatory index, decreased T-cell function
  - Chronic mucocutaneous disease
    - Absent proliferation to candida antigen in vivo
- Cytokine production: IL-2, IFN-Gamma
  - -IL-12R deficiency: decreased IFN-Gamma release
- TCR excision assay(TREC): circular DNA found in T-cells
  - -SCID

Lymphocyte Enumeration: **B-cells**

- Check Levels for CD19+, CD20+

-X-linked Brutons Agammaglobulinemia: Male predominant

 -No B-cell maturation: BTK gene

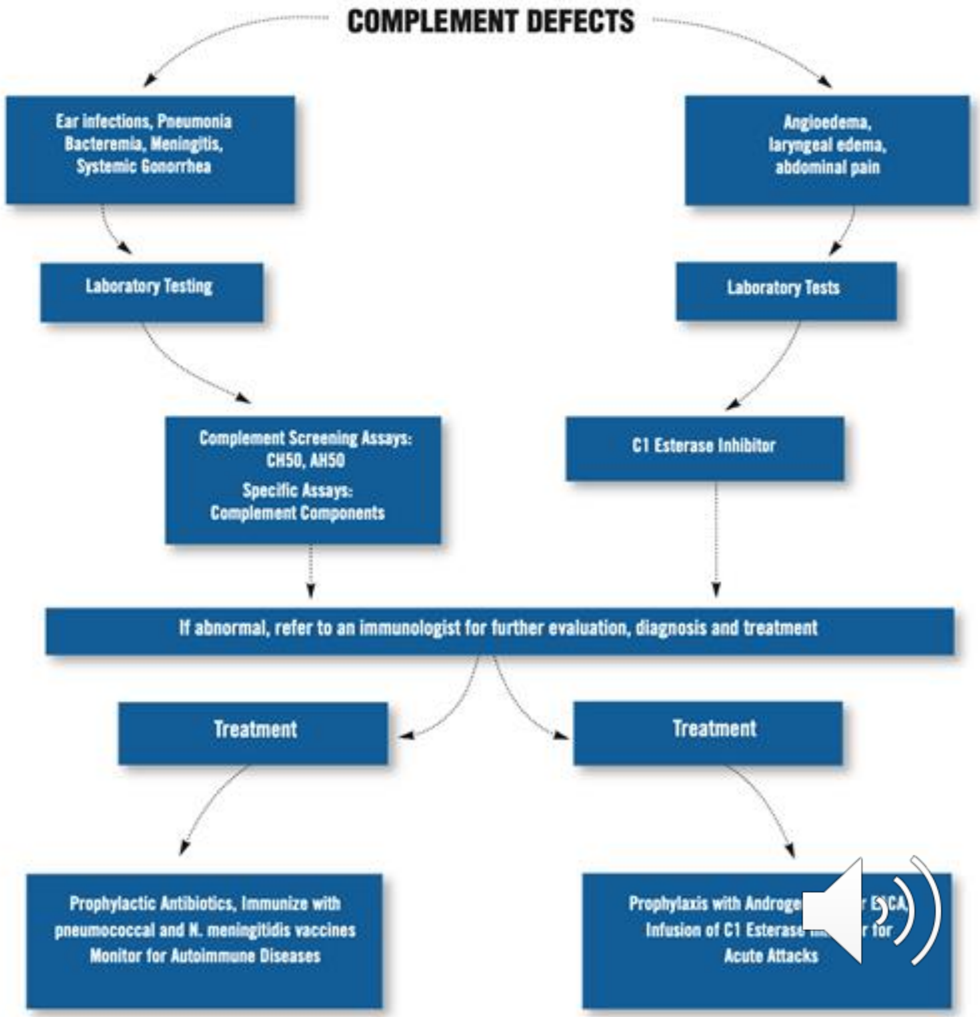
-SCID: T, B, & NK cells

-Ataxia Telangiectasia: ATM gene

-Pre-B Acute Lymphoblastic leukemia: Increased B-cells

Messenger Protein Abnormalities

1. Complement
2. Interleukins
3. Interferon



(Immune Deficiency Foundation, 2015)

Complement Quantity defect

- **CH50**: Classical pathway, Hemolytic complement assay
  - -non-specific, screening
- **AH50:** Alternative pathway: Hemolytic complement assay
  - -Neisserial infection, most common indication
  - -Determines deficiency in C3, C5-9
- **C1-C9 Enumeration**
  - **SLE**: C1-C4 ↓
  - ***Neisseria* Bacteremia**: C5-C9 ↓

Complement Disorders

C1-Esterase inhibitor deficiency:
C4 ↓
-Hereditary angioedema

Further importance on history taking and physical examination

Complement Disorders

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

- **Consumption:**
  - -C3,C4
  - -Lupus Nephritis

- **Deposition**: due to Ag/Ab complexes
  - -C3, C4, & C1q
  - -Post Streptococcal Glomerulonephritis: C3 ↓
  - -MPGN: C3 ↓

Interleukins

- **IL-1a/1b**
  -IBD, RA, & Psoriasis
- **IL-4**
- **IL-5**
  -Allergic Asthma
- **IL-6/12**
  -IBD, RA, SLE, & Asthma
  -Castleman disease
- **IL-7/9**
  -Autoimmune diseases
- **IL10**

Actemra/Tocilizumab

- MAB, monoclonal antibody
- IL-6 receptor inhibitor
- Used in treatment of inflammatory and autoimmune conditions
- Different types of arthritis
- COVID-19 use
- Manifests in alteration of the cytokine release pathways

Interferon Gamma Release Assay

- **Used to determine TB infection**

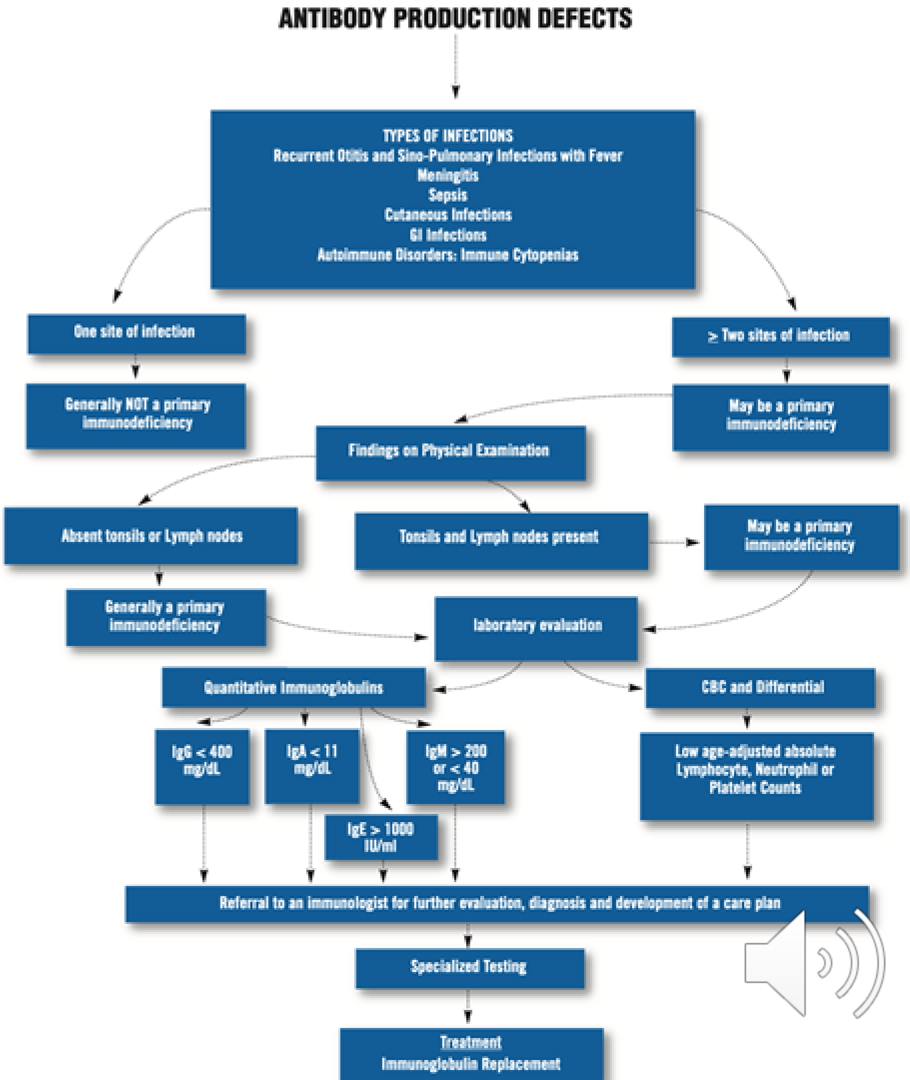
- **IGRA/QuantiFERON:** WBC release IFN-Gamma
  -Antigen-WBC mix
  -Levels tested for amount of IFN-Gamma released
  -Response: Determinate of LTB infection
  -Lab should reveal both qualitative & quantitative response

Ab Defects

- **Deficiency**:
  - Sinus and Pulmonary infections
  - Give way to infection by:
    - -*Streptococcus Pneumoniae*
    - -*Haemophilus Influenzae*
    - -Other viral agents

- **Overproduction**:
  - Multiple myeloma
  - MGUS
  - Waldenstrom Macroglobulinemia

(Immune Deficiency Foundation, 2015)



Quantitative Evaluation

- **IgG** in response to an Ag
  - -IS detects Ag
  - -B-cells function
  - -Invalid if patient received Ig in past 6 months

- **IgA**: IgA deficiency an extremely common cause of immune deficiency.
  - -Seen in recurrent mucosal infections, breakdown in initial barrier to infection (MALT)

- **IgM:** Brutons agammaglobunemia

- **IgE:** Hyper-IgE (Job) syndrome & Type I HSN

Quantitative Evaluation

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

- Wiskott-Aldrich syndrome: ↑IgA, ↑IgE; ↓IgG, ↓IgM
- AIDS: Increase in all Ig classes
- Ataxia-Telangiectasia: ↓IgA, IgE, IgG
- SLE: Increase in all Ig classes
- RA: Increase in IgA or all
- Scleroderma: increase in all Ig
- Celiac Disease: Low IgA
- Primary biliary cholangitis: ↑IgM

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

Levels of immunoglobulins in sera of normal subjects by age*

Age	IgG (mg/dL)	IgM (mg/dL)	IgA (mg/dL)	Total immunoglobulin (mg/dL)
Newborn	1031±200¶	11±5	2±3	1044±201
1 to 3 months	430±119	30±11	21±13	481±127
4 to 6 months	427±186	43±17	28±18	498±204
7 to 12 months	661±219	54±23	37±18	752±242
13 to 24 months	762±209	58±23	50±24	870±258
25 to 36 months	892±183	61±19	71±37	1024±205
3 to 5 years	929±228	56±18	93±27	1078±245
6 to 8 years	923±256	65±25	124±45	1112±293
9 to 11 years	1124±235	79±33	131±60	1334±254
12 to 16 years	946±124	59±20	148±63	1153±169
Adults	1158±305	99±27	200±61	1457±353

IgG: immunoglobulin G; IgM: immunoglobulin M; IgA: immunoglobulin.

* The values were divided from measurements made in 296 healthy children and adults. Levels were determined by the radial diffusion technique using specific rabbit antisera to human immunoglobulins.

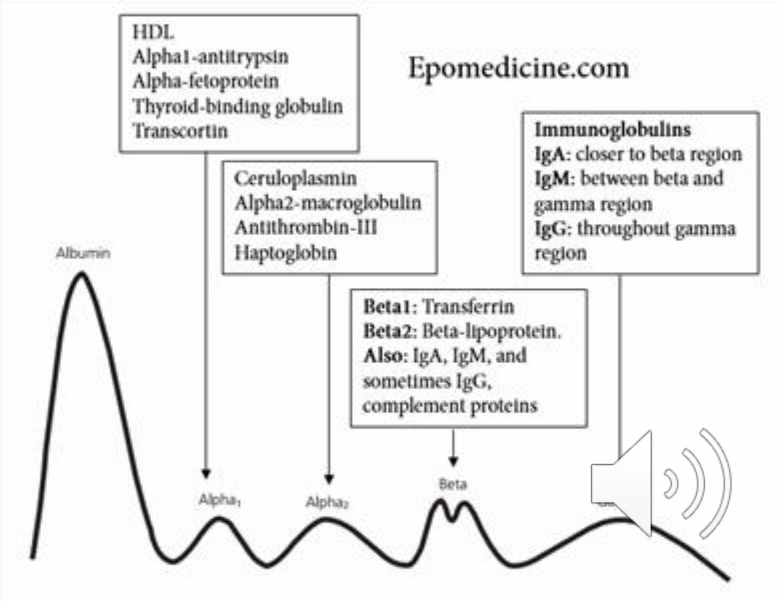
¶ One standard deviation.

Graphic 110584 Version 4.0

Quantitative Evaluation

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

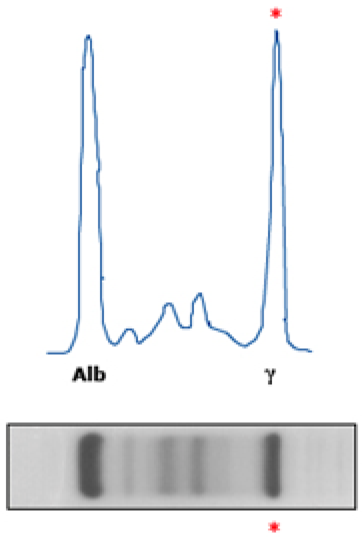
- **Serum Protein Electrophoresis: SPEP**
  - -M Protein of all Igs
    - -Abnormal protein made that leads to pathology
- **Important in diagnosing:**
  - -Multiple myeloma: IgG/IgA M-protein
  - -MGUS:
    - <10% IgM M protein in Bone Marrow
  - -Waldenstrom Macroglobulinemia:
    - >10% IgM M protein



Monoclonal pattern on serum protein electrophoresis (SPEP)

A

B



(A) Densitometer tracing of these findings reveals a tall, narrow-based peak (asterisk) of gamma mobility and has been likened to a church spire. The monoclonal band has a densitometric appearance similar to that of albumin (alb) and a reduction in the normal polyclonal gamma band.
(B) A dense, localized band (asterisk) representing a monoclonal protein of gamma mobility is seen on SPEP on agarose gel (anode on left).

Type 1 Hypersensitivities

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

- **IgE** reaction to external Ag
  - -Specific to Ag presented
  - -Can lead to anaphylaxis
  - -Increased level seen in Atopic patients
  - -Involved in type I HSN
- **Skin Testing**

- **Exogenous/Environmental**: Food, pollen, molds, animal dander, & insects

Autoimmune

Anti-Nuclear Antibody: ANA
Anti-dsDNA/Anti-Sm: SLE
Anti-MPO/P-ANCA: UC
Anti-U1 RNP: MCTD
Anti-SSA/SSB: Sjögren
Anti-Scl70/Anti-DNA Topoisomerase: Scleroderma
Anti-Cardiolipin: SLE, Anti-phospholipid syndrome
Anti-CCP:RA
Anti-CD55/DAF: Paroxysmal nocturnal hemoglobinuria
Anti-Thyroid peroxidase: Hashimoto Thyroiditis
Anti-Parietal Cells: Pernicious anemia
Anti-Endomysial: Celiac disease
Anti-Smooth Muscle: Autoimmune hepatitis type 1
Anti-Mitochondiral: Primary Biliary cholangitis

Vasculitis

- Cryoglobulins: Rheumatoid Factor
  - -Rheumatoid Vasculitis

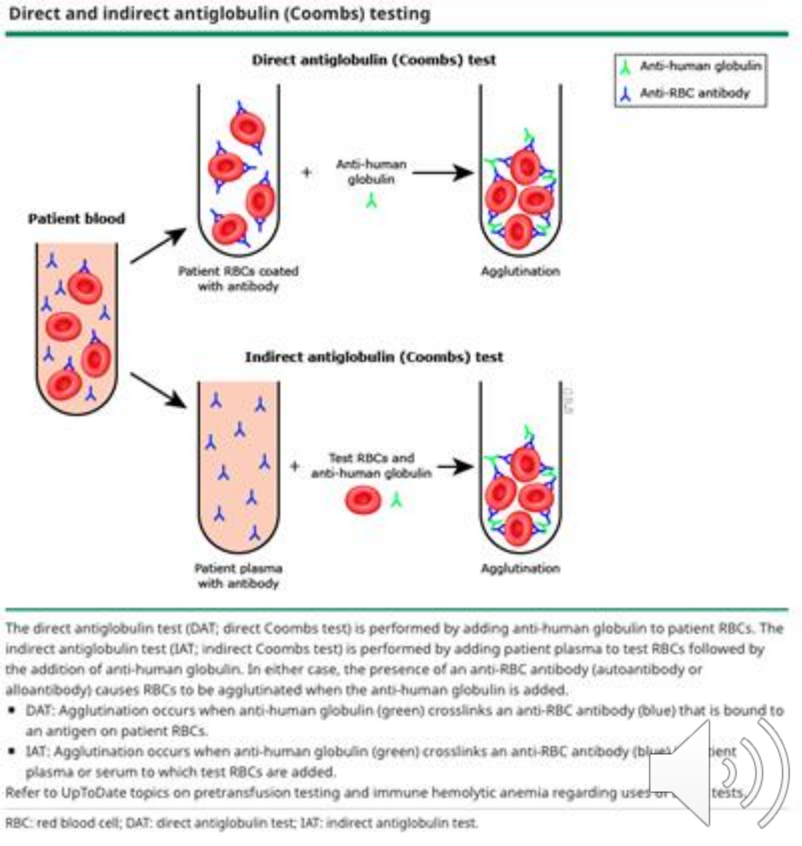
- ANCA mediated:
  - -C-ANCA: Wegner's Granulomatosis
  - -P-ANCA: Microscopic Polyangiitis
- Others:

a) **Rheumatoid vasculitis: RF**
b) **Cutaneous small vessel vasculitis**
c) **IgA Vasculitis**
d) **MPA and Churg strauss**
e) **Wegner/GPA**

Direct/Indirect Coombs test

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

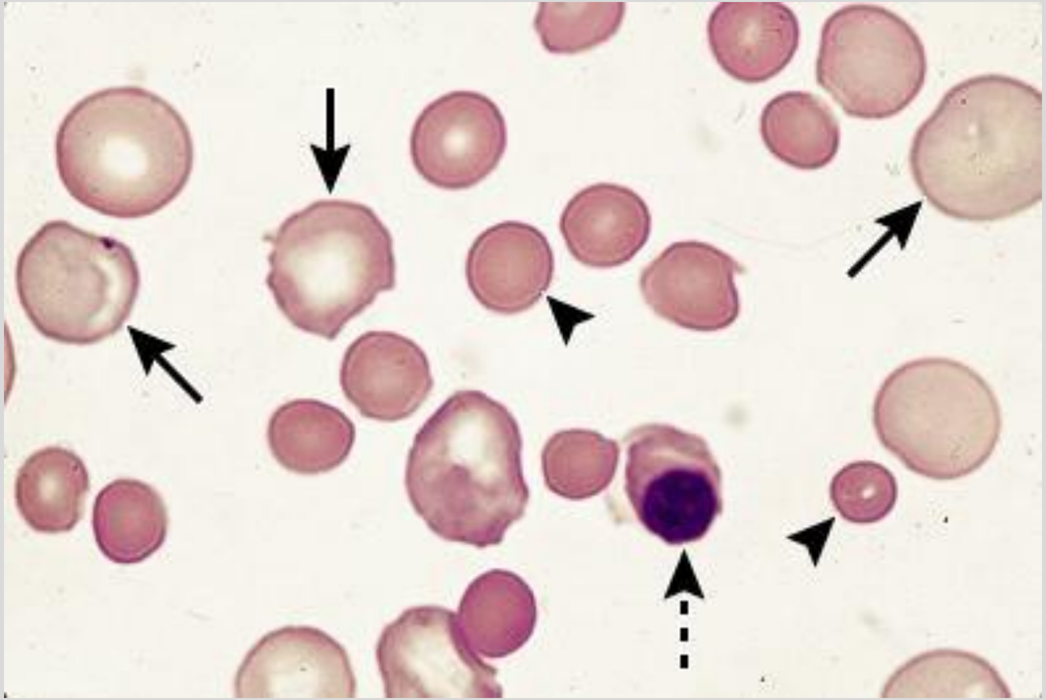
- Determine source of hemolysis due to autoimmune source
- Anemia, pretransfusion, suspected autoimmune hemolysis
- **Direct:** Check for Ab/RBC complex
- **Indirect:** Check for Abs for RBC in plasma



(Brugnara, et al., 2021)

Blood Smear

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine



Coombs-Positive
Autoimmune hemolytic anemia

1. Spherocytes
2. Nucleated RBC
3. Polychromatophilic Red Cells

Tuberculin Test

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

- Delayed/Type IV hypersensitivity
  - -Intradermal injection with Ag: PPD
  - -Requires **APC** & **CD4+ cells**
  - -Not valid in those who have compromised cellular immunity
  - -Wait **48-72 hours** after injections to evaluate response
  - \- Positive test - latent infection/<u>Previous Vaccination</u>

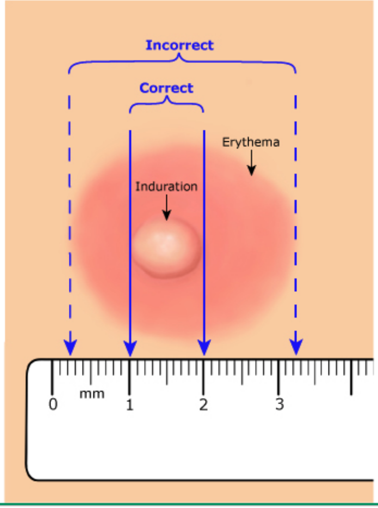
A comparison of tuberculin skin tests and interferon-gamma release assays

	TST	IGRAs
Advantages	TST does not require equipment and can be done without a laboratory. It is less expensive than for IGRAs (but greater personnel time is required). Longitudinal studies have demonstrated the predictive value of TST, and randomized trials show that LTBI therapy is much more effective in those who are TST positive than TST negative.	IGRA does not require follow-up visit to complete the testing process (but follow-up visits may be needed for LTBI therapy). IGRA does not give false-positive results because of prior BCG vaccination.
Disadvantages	TST requires an intra-dermal injection and a subsequent follow-up visit with trained staff to interpret results in 48 to 72 hours. TST may give false-positive results because of prior BCG vaccination or sensitization to nontuberculous mycobacteria. TST may give false-negative results because of immunosuppression (eg, HIV or active TB), natural waning of immunity, and technical limitations (including reader variability). Serial TST interpretation is complicated by potential boosting, although this can be minimized through initial two-step testing. TST can cause adverse reactions (skin blistering and ulceration); these are rare.	IGRA requires a blood draw, laboratory equipment, and technical expertise for specimen collection, processing, and assay. Reagent costs are substantially higher than costs of TST. IGRA test results are available within 24 hours at the earliest. IGRA may give false-negative results because of immunosuppression (eg, HIV or active TB) and technical variability. Interpretation of serial IGRA is complicated by frequent conversions and spontaneous reversions and by lack of consensus on optimum thresholds for conversions.

TST: tuberculin skin test; IGRA: interferon-gamma release assay; HIV: human immunodeficiency virus; LTBI: latent tuberculosis infection; BCG: Bacillus Calmette-Guerin.

Graphic 76107 Version 4.0

Measuring a reaction to the tuberculin skin test



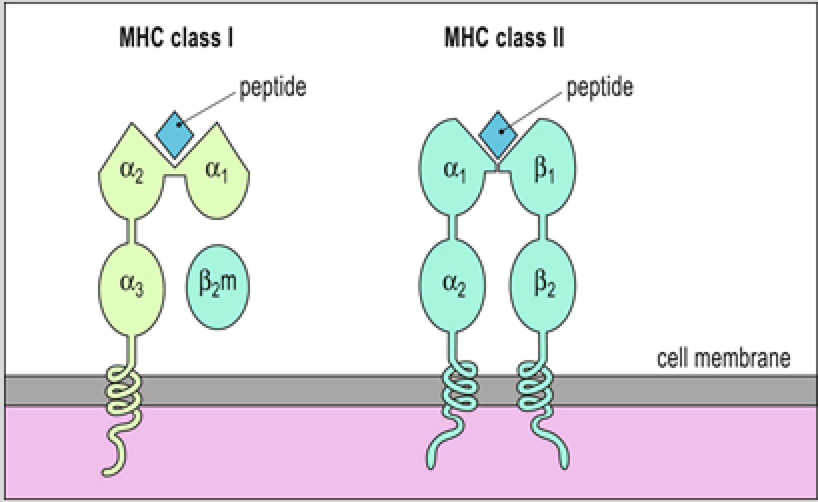
This figure shows the correct method for measuring a reaction to the tuberculin skin test. The size of the reaction is measured by the width of induration, not erythema. In the example shown, the reaction measures 10 mm.

Redrawn from: Testing for Tuberculosis Infection and Disease. In: Core Curriculum on Tuberculosis: What the Clinician Should Know, Sixth Edition, Centers for Disease Control and Prevention, 2013. Available at: https://www.cdc.gov/tb/education/corecurr/index.htm.

Graphic 80766 Version 5.0

Human Leukocyte Antigen

- **MHC-I HLA**: Self Ag
  - -HLA-A, B, & C
  - -Important in transplantation
  - -HLA-DR2: MS, SLE, Goodpasture
  - -HLA-DR3: DM-I, SLE, Graves
  - -HLA-DR4: RA, DM-1, Addison's
- **MHC-II HLA:** Foreign Ag
  - -HLA-DR, DQ, & DP
  - -Associated with autoimmune:
    - HLA-B27: Psoriatic arthritis, Ankylosing Spondylitis, IBD-associated arthritis, reactive arthritis



(Male, et al., 2021)

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

Lecture Feedback Form:

https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWYXREL4R

Shil Punatar
shilpunatar@gmail.com

References

NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic Medicine

- American Board of Internal Medicine. (2020). *ABIM Laboratory Reference Ranges* [PDF]. Philadelphia: American Board of Internal Meidince.

- Brugnara, C., Dr., & Brodsky, R., Dr. (2021, January). Warm Autoimmune Hemolytic Anemia in Adults. Retrieved February 10, 2021, from https://www.uptodate.com/contents/warm-autoimmune-hemolytic-anemia-aiha-in-adults?search=coombs+test&source=search_result&selectedTitle=1~150&usage_type=default&display_rank=1

- Buckley, R., Dr., Ballow, M., Dr., Fleisher, T., Dr., Rosenzweig, S. D., Dr., Blaese, R. M., Dr., Holland, S., Dr., . . . Winklestein, J., Dr. (2015). *Diagnostic and Clinical Care Guidelines* [PDF]. Townson, MD: Immune Deficiency Foundation.

- Butte, M. J., Dr., & Stiehm, E. R., Dr. (n.d.). Laboratory evaluation of the Immune System-UpToDate. Retrieved February 01, 2021, from https://www.uptodate.com/contents/laboratory-evaluation-of-the-immune-system?source=autocomplete&index=0~4&search=labratory

- Epomedicine. (2016, October 24). M-spike. Retrieved February 1, 2021, from https://epomedicine.com/medical-students/m-spike/

References

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

- Kushner, I., Dr. (2021, January). Acute Phase Reactants. Retrieved February 1, 2021, from https://www.uptodate.com/contents/acute-phase-reactants?search=acute+phase+reactants&source=search_result&selectedTitle=1~150&usage_type=default&display_rank=1
- Liu, Lucy. "Chronic Granulomatous Disease." *Chronic Granulomatous Disease - Immunology - Medbullets Step 1*, 30 Nov. 2020, step1.medbullets.com/immunology/105009/chronic-granulomatous-disease.
- Male, D. K., Peebles, S., & Male, V. (2021). *Immunology*. Philadelphia, PA: Elsevier.
- Menzies, D., Dr. (2021, January). Use of Interferon-Gamma Release Assay for diagnosis of Latent Tuberculosis infection in adults. Retrieved February 10, 2021, from https://www.uptodate.com/contents/use-of-interferon-gamma-release-assays-for-diagnosis-of-latent-tuberculosis-infection-tuberculosis-screening-in-adults?search=igra&source=search_result&selectedTitle=1~137&usage_type=default&display_rank=1
- Menzies, D., Dr. (2021, January). Use of Tuberculin Skin Test for Diagnosis of Latent Tuberculosis infection in Adults. Retrieved February 1, 2021, from https://www.uptodate.com/contents/use-of-the-tuberculin-skin-test-for-diagnosis-of-latent-tuberculosis-infection-tuberculosis-screening-in-adults?search=PPD&source=search_result&selectedTitle=1~44&usage_type=default&display_rank=1

References

NEW YORK INSTITUTE OF TECHNOLOGY
College of Osteopathic Medicine

- Rajkumar, S. V., Dr. (2021, January). Laborattory methods for analyzing monoclonal proteins. Retrieved February 1, 2021, from https://www.uptodate.com/contents/laboratory-methods-for-analyzing-monoclonal-proteins?search=spep&source=search_result&selectedTitle=1~134&usage_type=default&display_rank=1

- Rosenthal, D. S., Dr. (2021, January). Evaluation of The Peripheral Blood Smear. Retrieved February 1, 2021, from https://www.uptodate.com/contents/evaluation-of-the-peripheral-blood-smear?search=blood+smear&source=search_result&selectedTitle=1~150&usage_type=default&display_rank=1

- Ryan, J., Dr. (2021, January). Boards and Beyond: Immunology. Retrieved February 1, 2021, from https://boardsbeyond.com/category/step-1/immunology

- Sax, P. E., Dr. (2021, January). Screening and Diagnostic testing for HIV infection. Retrieved February 1, 2021, from https://www.uptodate.com/contents/screening-and-diagnostic-testing-for-hiv-infection?search=hiv+testing&source=search_result&selectedTitle=1~150&usage_type=default&display_rank=1

- Staff, H. (2019, February). Complete blood count (cbc). Retrieved February 10, 2021, from https://www.mottchildren.org/health-library/hw4260#hw4305

- Vikas, L. T., Bhushan, V. M., Sochat, M. M., Kallianor, K. M., Abrams, J., & Baidyanathan, V. M. (2020). Microbiology and Immunology. Hematology and Oncology. In *First aid for the USMLE step 1 2019: A student-to-student guide* (pp. 95-395). New York,, NY: McGraw Hill Education.